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Impact of Audible Sound on *In vitro* Growth of Algae

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ABSTRACT

Algae are represented by unicellular or multicellular autotrophic thalloid group of plants that are cosmopolitan in occurrence. To study the impact of audible sound on algal growth and physiology four algal specimens were selected for experiment namely *Microcystis* sp., *Arthrospira* sp., *Chlorococcum* sp., *Cladophora* sp. Three different frequencies were applied viz 432 Hz, 1000 Hz, 2500 Hz and with sound intensity of 30 dB and 60 dB. Selected sound frequencies showed positive effect on the *in vitro* algal growth along with influencing the biochemical parameters of the treated algae.

Key words: *Audible sound, Growth, frequency,* In vitro, *algae*

Introduction

Algae are the lower group of autotrophs, thalloid, unicellular or multicellular in organisation that are found in diverse habitats like fresh water, sea water, brackish water, in land, like soil, hill area, ice zones etc. Cyanobacteria, also known as blue green algae, are the group of bacteria that are capable to photosynthesis (Allaby, 1992). Algae are very useful to both ecological and economical field. Algae can provide vitamins, potassium, magnesium, calcium, and iodine, with great nutritional value (Morton and Steve, 2008). It able to capture different polluting agent and help to filtered polluted water (Morrissey *et al.*, 2015). *Microcystis* sp. is blue green algae which belong to the family Microcystaceae. It is also involved in production of biodiesel (Ashokkumar *et al.,* 2014). *Arthrospira* sp. is the filamentous blue green alga which belongs to Microcoleaceae family. It possesses great nutritional value owing to its high protein concentration which can reach upto 51-71% of its biomass (Companellal *et al*., 2002). The unicellular green alga *Chlorococcum* sp. of the family Chlorococcaceae is abundantly found in the fresh water, soil, tree barks, stones etc. (Shubert, 2014). It has ability to huge lipid production. *Cladophora* sp. is represented by branched filamentous thallus belonging to the family of Cladophoraceae. It is abundantly used as a source of bio-fuel (Michalak and Messyasz, 2021). It also shows enough potency as antimicrobial, antioxidant and anti-inflammatory activities. Due to these reason it shows great pharmaceutical activity.

Sound after being generated from any sound source, can be transmitted in any medium (Walters, 1969). Sound can be of various kinds based on frequencies, among which the frequency of audible sound ranges from 20Hz to 20 kHz. Environmental conditions can place a biological system under stress condition which can affect its physical growth and other biochemical parameters. The *in vitro* growth of algae *Haematococcus pluvialis* were studied in presence of audible sound for 8 hours daily up to a period of 22 days (Hadiyanto and Christwardana,

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2017). After the stipulated time it was found that the algal growth increased up to 58% as compared to that of control. Similar results were also reported by working on *Chromobacterium violaceum* where it's *In vitro* growth increased up to 8.1% when it was treated with audible sound from 41Hz to 645 Hz frequency (Kothari and Sarvaiya, 2017).

Materials and Methods

Algal samples were collected from different regions of Malda district, West Bengal, India. Then *In vitro* pure culture of *Microcystis* sp., *Arthrospira* sp., *Chlorococcum* sp., *Cladophora* sp. was established using BG-11 culture medium. In this experiment the *in vitro* algal growth rate were subjected to different frequencies, different loudness and time. Sound frequency ranged from 432 Hz and 2500 Hz sound frequency, loudness ranges from 30 dB and 60 dB loudness for 2 hours and 4 hours duration daily. The growth rate of algae were measured at an interval of three days interval using spectrophotometric method measuring absorbance at 680 nm wavelength. Biochemical parameters of the algal genera under study were also measured. Amount of protein estimated by the using Bradford method (1976). Anthrone method (1946) was used to estimate the carbohydrate of selected algal samples. Lipid content was estimated by using Vaniline reagent. Chlorophyll a of the selected algal samples were also measured as per Arnon, 1949. All the statistical data were analysed in SPSS v. 24.0 (n=3, p=0.05), such trend was also followed in related investigations

Fig. 1. *In vitro* experimental setup

Results and Discussion

From this experiment it was found that, the audible sound showed significant role in growth of algae. After using audible sound treatment on selected algae it was notified that the audible sound showed positive impact on the growth of algae. The unicellular blue green alga *Microcystis* sp. treated with audible sound (432 Hz, 60 dB) up to 2 hours daily it showed 34.1 % higher growth than the control set. Similar results were obtained by using audible sound applied on filamentous blue green alga *Arthrospira* sp. Here the audible sound frequency were 432 Hz frequency with sound intensity of 60 dB applied up to 4 hours daily that resulted in 41.7 % higher growth than the control set. The green alga *Chlorococcum* sp. cultured using audible sound (2500 Hz, 60 dB) 4 hours daily showed 47.5% higher growth rate than the control set. The branched filamentous green alga *Cladophora* sp. also treated with audible sound (2500 Hz, 60 dB) 4 hours daily and as a result it was found that the growth of this alga increase 17.4 % higher than the control set.

From this experiment it is evident that algae can increase its growth in presence of audible sound with particular frequency and loudness. Bio-chemicals (Protein, carbohydrate, chlorophyll-a, lipid) estimation done on those experimental algae, i.e. the sound treated algae and the untreated algae. After estimation of biochemical profile on both treated and untreated algae it was found that the audible sound showed a positive impact on the biochemical contents of the sound treated algae in comparison to the control set ups. When *Microcystis* sp. (S1) was treated with audible sound, its protein content increased 21.80 %, carbohydrate content increased 17.20 % and chlorophyll-a content increased 3.68 % and lipid content increased 14.2% than the control set up. The blue green filamentous algae *Arthrospira* sp. (S2) treated with audible sound resulted in 29.93% increase in protein content, whereas carbohydrate increased by 17.81%, chlorophyll-a increased by 4.78% and lipid increased 11.5% in comparison to control culture. The green microalgae *Chlorococcum* sp. (S3) increased its protein content 44.83 %, carbohydrate content 16.28%, chlorophyll-a content 6.40% and lipid content 22.8% in presence of audible sound. *Cladophora* sp. (S4) treated with audible sound resulted in the protein content increase of 10.89%, carbohydrate content increased by 11.80%, chlorophyll-a content increased by 4.43% and lipid increased up to 5.5% than the untreated cultures of *Cladophora* sp.

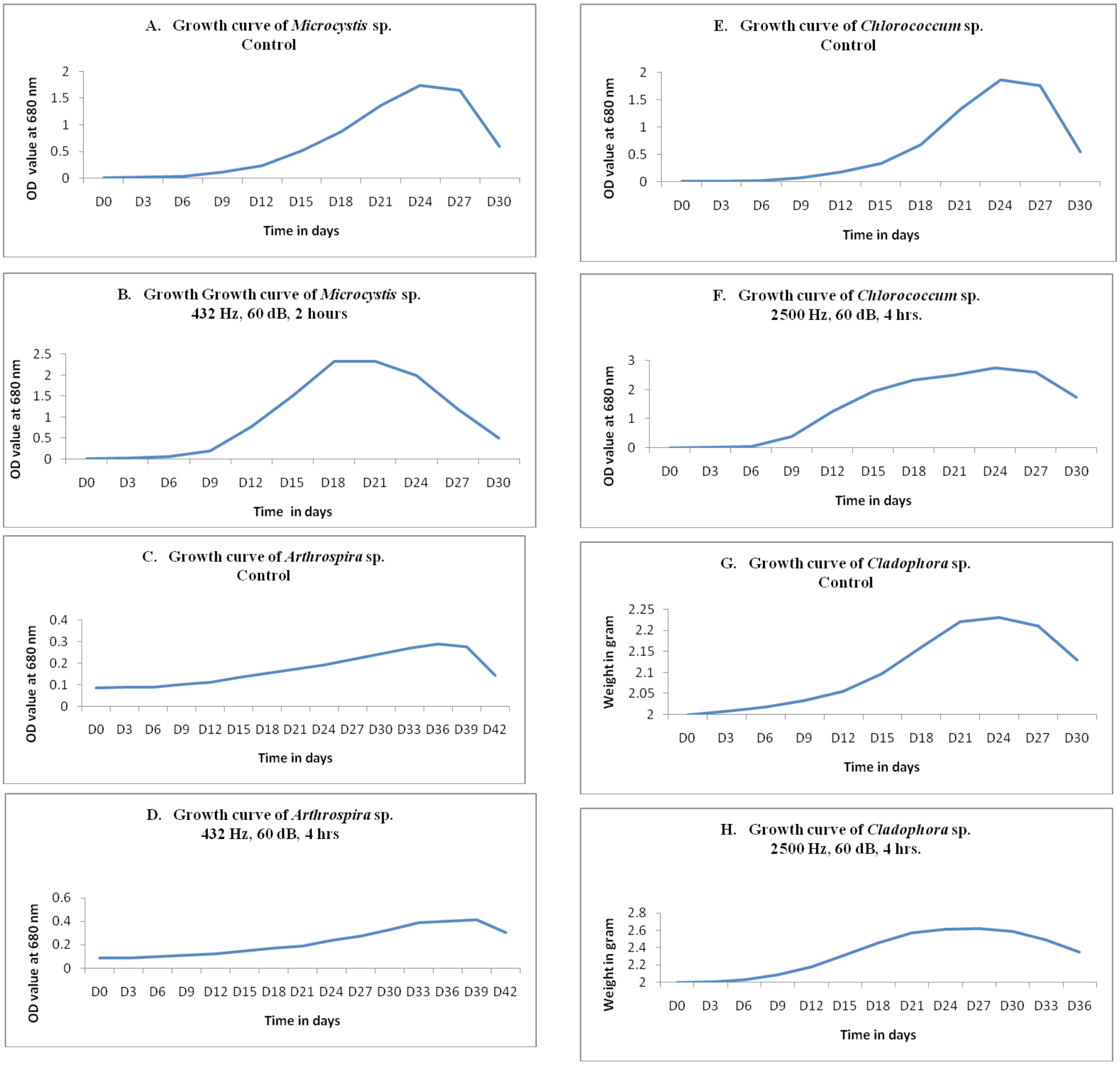

Fig. 2. A., C., E., G. – Growth curve of control culture and B., D., F., H. - Audible sound treated algae (*Microcystis* sp., *Arthrospira* sp., *Chlorococcum* sp., *Cladophora* sp.) respectively.

Conclusion

Different algal specimen was collected from different blocks of Malda district, West Bengal, India. Out of these most abundant algae all over the Malda were selected as the experimental algae. This selected unicellular and filamentous algae belongs to both blue green and green algal group. Increasing growth of algae using such kind of physico-stimulant is an ecofriendly method and also support production of huge algal biomass within budget friendly. Previous researches have indicated that the application of various nature of audible sound as stimuli can promote the growth of different plants. If plant exposed to audible sound containing with certain frequency and intensity, it helps to promotion in growth, resistance to parasites and resistance to diseases also. Depending upon the sound frequency and intensity, plant can promote its growth (Hassanien *et al*., 2014).

The green bean exposed to audible sound with frequency 1-2.5 kHz and 80-100 dB sound intensity

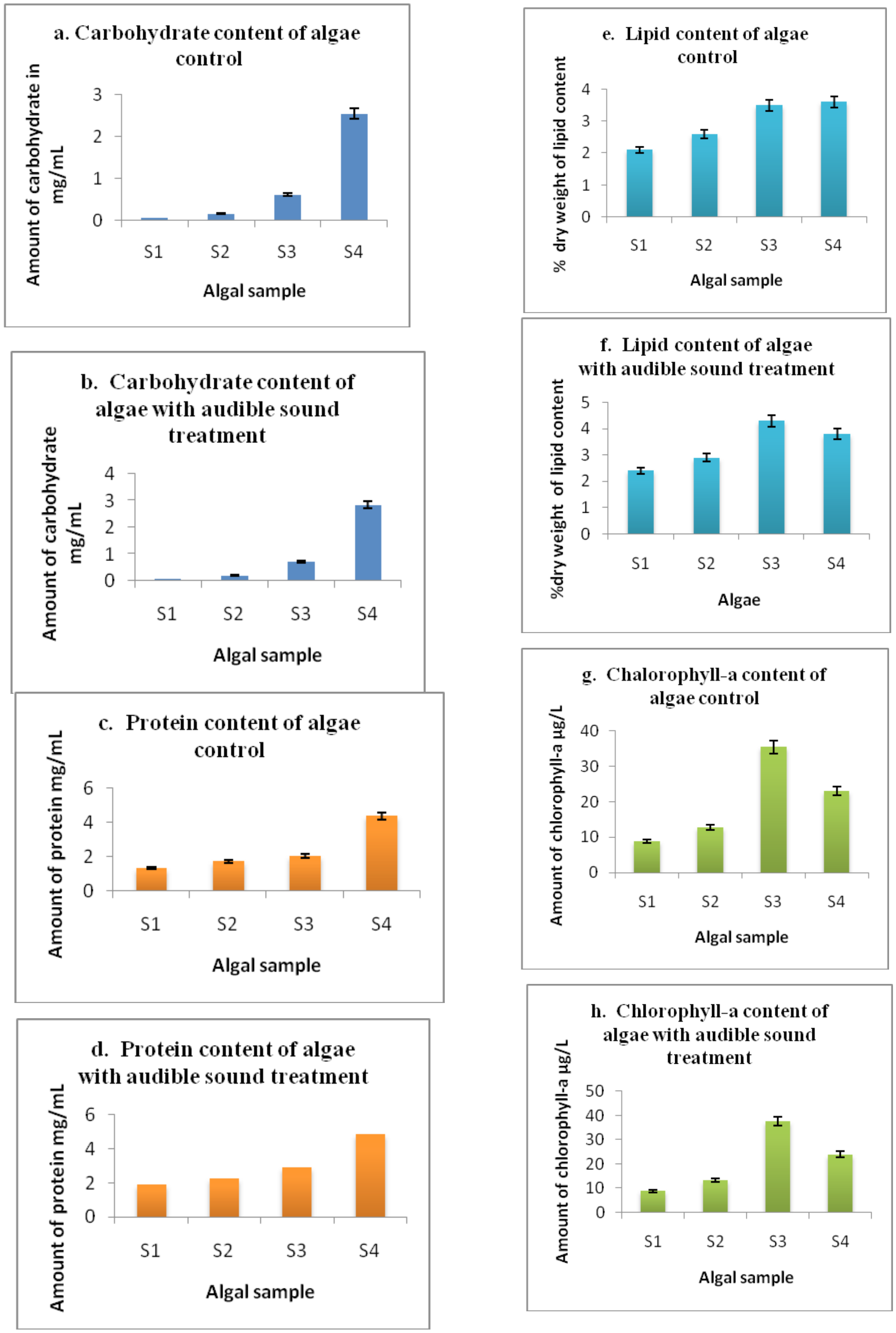

Fig. 3. Graphical representation of Carbohydrate content (control culture-**a**.and treated culture -**b**.), Protein content (control culture-**c**. and treated culture-**d**.), Lipid content(control culture-**e**.and treated culture-**f**.), chlorophyll-a content(control culture and **g**. and treated culture-**h**.) in algae. [Here, S1-*Microcystis* sp., S2- *Arthrospira* sp., S3- *Chlorococcum* sp., S4- *Cladophora* sp.]

for 72 hours showed positive results. Here the period of seed germination become decreased (Cai *et al.,* 2014). When tomato plants were treated with different sound frequencies like 6 kHz, 1.24 kHz 1.6 kHz with 90 dB intensity, the phenol content of treated plant was increased up to 70 %, ascorbic acid content increased up to 14 % and Lycopene content increased up to 20 %. Here the sound frequency 1.6 kHz showed best performance to increased total sugar, total acids, phenol vitamin-C, lycopene also (Altunatas and Ozkurt, 2019). Yeast cells exposed to audible sound with different set ups. It was treated with 100 Hz frequency and 90 dB sound intensity, broadband 320 kbps at 80/20 dB sound intensity. After this experiment, as a result it was found that the growth rate of yeast cells increased up to 12 %. Different sound frequencies also treated on *Escherichia coli* strain (Aggio *et al.,* 2012). If plant exposed to medium or low audible sound frequency and intensity, the treatment increased the growth rate, photosynthetic rate and pest resistance (Hassainen *et al.,* 2014 and Ghosh *et al*., 2016). Thus impact of audible sound has been primarily proved although the pathway of growth promotion is not elucidated. The impact of sound especially audible sound on algal biomass is a relatively less explored field of study, however very recently some research works have been initiated but the exact reason or the cascading process of sound - mediated growth promotion or inhibition (in case of ultra sound) is still not clearly understood.

Conflict of Interest - None

References

Allaby, M. 1992. Algae. *The Concise Dictionart of Botany*. Oxford: Oxford University Press.

Altuntas, O. and Ozkurt, H. 2019. The assessment of tomato fruit quality parameters under different sound waves. *J Food Sci Technol.* 56(4): 2186-2194.

Ashokkumar, V.,Agila, E., Salam, Z.,Ponraj, M., Md Din, M.F. and Ani, F.N. 2014. A study on large scale cultivation of *Microcystisaeruginosa* under open raceway pond at semi-continuous mode for biodiesel production. *Bioresour. Technol*, pp. 186-193.

Banerjee, S., Goswami, A., Datta, A., Pyne, A., Nikhat, A. and Ghosh, B. 2018. Effect of Different Sound Frequencies on the Growth and Antibiotic Susceptibility of *Escherichia coli*. *Int J Curr Microbiol Appl Sci*. 7(3): 1931-1939.

Campanella, L., Russo, M.V. and Avino, P. 2002. Free and total amino acid composition in blue-green algae. *Annali di Chimica.* 92(4): 343-352. PMID 12073880.

Michalak, I. and Messyasz, B. 2021. Concise review of Cladophora spp.: macroalgae of commercial interest. *J Appl Phycol.* 33:133-166.

Morrissey, J., Jones, M.S. and Harriott, V. 2015. Nutrient cycling in the Great Barrier Reef Aquarium- *Proceedings of the 6th International Coral Reef Symposium*, Australia. ReefBase.